

PARTH SINGH

📍 Bangalore

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🌐 Portfolio

Full-stack engineer with experience in scalable web applications, **backend APIs**, and **GenAI integrations**. Skilled in **Next.js**, **Node.js**, and **AI-powered systems** with a focus on performance and practical problem-solving.

EXPERIENCE

IBM | Associate System Engineer

May 2026 - Present

- Undergoing enterprise training in SAP SCM MM and business process workflows.
- Building technical depth alongside enterprise systems exposure.

Bytemasks - Remote | Frontend Developer Intern

Mar 2025 - July 2025

- Built production-ready dashboard UI for **Monitor** using Next.js and TypeScript with real-time notifications.
- Expanded **Safe AI** by developing Lab, Collaboration, and Prompt Library features to improve user workflow.
- Developed FastAPI endpoints (bookmarks, sharing) and optimized backend performance using Redis caching and request debouncing, reducing repeated DB queries by ~35%.
- Contributed to production deployments in an agile workflow with structured GitHub version control and cross-team coordination.

PROJECTS

RAG-Based Internal Knowledge Assistant - (Next.js, TypeScript, Pinecone, Gemini, Grok, Langchain)

[Live Demo]

- Built an enterprise-style **Retrieval-Augmented Generation (RAG)** system grounded strictly on internal documents, preventing hallucinated responses.
- Designed secure **document ingestion pipeline** using **LangChain** PDFLoader and semantic chunking (500 size, 100 overlap) with **3072-dimension Gemini embeddings** stored in **Pinecone**.
- Implemented **cosine-similarity retrieval (topK=3)** and constructed contextual prompts for **Groq LLaMA 3.1 (temperature=0)** to ensure deterministic, context-bound answers.
- Deployed **backend-only vector querying** architecture on Vercel with clear separation between ingestion, retrieval, and generation layers.

MINDSCRIBE – (Next.js, Node.js, Express.js, MongoDB, Cloudinary, JWT, Redis, Gemini AI, Hugging Face)

[Live Demo]

- Built a scalable blog management system with AI-assisted content workflows and modular backend architecture.
- Integrated **Gemini** and **Hugging Face** APIs for summarization, title generation, and grammar enhancement using prompt-based requests.
- Implemented **JWT authentication**, **pagination**, **debounced search** with **backend filtering**, and Cloudinary-based media handling.
- Optimized **read-heavy APIs** using **Redis caching** and **rate limiting**, improving response times by ~25–30% under repeated requests.

SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL
Frontend	React.js, Next.js, Redux Toolkit, Tailwind CSS
Backend	Node.js, Express.js, FastAPI, REST APIs
Database	PostgreSQL, MongoDB, Redis, SQLAlchemy ORM
AI/GenAI	LangChain, LLM Integrations, RAG Pipelines, Prompt Engineering, Vector Databases, AI API Integrations
Tools	Git, GitHub, Docker, Postman
Core Concepts	Data Structures & Algorithms, DBMS, OOPs, Authentication (JWT), API Design

EDUCATION

B.Tech in Computer Science & Engineering –

2021 – 2025 | CGPA: 7.19

AKTU University | Maharana Pratap Engineering College

ACHIEVEMENTS

- AIR 66 in MPGI Tech-A-Thon (2023) among 300+ participants.
- GATE 2026 – Qualified (Score: 396)